

From R_τ to the Second Electrical Revolution: Timeline and Technological Implications of Room-Temperature Superconductivity

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(Dated: March 28, 2026)

We combine the R_τ screening criterion for superconductivity—derived from the Petz recovery framework—with current experimental progress to construct a quantitative timeline for room-temperature superconductivity (RTSC). The analysis identifies four milestones: 0 °C superconductivity at ambient pressure (2027–2030), room-temperature (20 °C) new materials (2030–2035), first commercial products (2035–2045), and systemic technological transformation (2040–2060). For each RTSC-enabled technology—lossless power transmission, compact MRI, nuclear fusion, superconducting energy storage, frictionless transport, and low-power computing—we provide quantitative impact estimates grounded in physics. Global transmission losses of ~ 2000 TWh/year would be eliminated, saving $\sim \$200$ B/year and ~ 1.5 Gt CO₂/year; MRI costs would decrease from \$1.5–3 M to \$100–200 K per unit, enabling universal diagnostic access; fusion reactor size and cost could decrease by an order of magnitude. We also delineate what RTSC *cannot* improve—quantum computing coherence, battery chemistry, and optical communication—to temper overenthusiasm. The R_τ screener provides a community-accessible tool for accelerating the search.

INTRODUCTION

The pursuit of room-temperature superconductivity (RTSC)—zero electrical resistance at $T \geq 300$ K and ambient pressure—spans more than a century since Onnes’s discovery of superconductivity in mercury at 4.2 K [1]. Progress accelerated dramatically in recent years: the compressed hydride LaH₁₀ achieves $T_c = 260$ K = -13 °C at 170 GPa [4, 5], while Deng and Chu demonstrated that pressure-quenched HgBaCaCuO retains $T_c = 151$ K at ambient pressure [7]—the highest ambient-pressure T_c ever recorded.

Despite this empirical momentum, the field has lacked a *quantitative* framework for predicting which materials can achieve RTSC and on what timescale. The R_τ screening criterion [9], derived from the Petz recovery formalism [10, 11], provides exactly this:

$$R_\tau(T) \equiv \frac{I_{\text{pair}}(T)}{\tau_{\text{ph}}(T)} \geq 1 \implies \text{superconductivity at } T, \quad (1)$$

where I_{pair} is the total pairing information (summed over all channels) and τ_{ph} is the phonon-induced decoherence. The criterion is mechanism-independent and computable from materials database quantities [9].

In this paper, we leverage the R_τ framework together with current experimental data to construct a timeline for RTSC milestones (Sec.), provide quantitative impact estimates for six key technologies (Sec.), delineate what RTSC cannot achieve (Sec.), and outline a research roadmap (Sec.).

TIMELINE PREDICTIONS BASED ON R_τ

The following predictions are grounded in two inputs: (i) R_τ values computed for known and predicted materials in Ref. [9], and (ii) the demonstrated success of pressure-quench techniques [7]. Confidence levels reflect the number of independent assumptions required.

Prediction 1 (2027–2030): 0 °C at ambient pressure

LaH₁₀ at 170 GPa has $R_\tau = 3.47$, far exceeding unity [9]. Its $T_c = 260$ K = -13 °C is already within 13 K of the 0 °C threshold. The critical question is whether the $Fm\bar{3}m$ clathrate structure can be metastably retained at ambient pressure.

The pressure-quench approach demonstrated by Deng and Chu [7] for HgBaCaCuO—a cuprate with very different chemistry—establishes the viability of kinetic trapping. Applying the same technique to LaH₁₀: rapid decompression at $T < 20$ K exploits the negligible hydrogen diffusivity ($D_H \sim 10^{-30}$ cm²/s) to freeze the clathrate cage.

Even with partial structural decomposition reducing T_c by 30%, the R_τ analysis gives

$$R_\tau^{\text{quench}} \approx 0.7 \times 3.47 = 2.43 \gg 1, \quad (2)$$

yielding $T_c \sim 180$ – 230 K ≈ -90 to -40 °C in the pessimistic case, and $T_c \sim 240$ – 260 K ≈ -30 to -13 °C if the structure is largely preserved. Encapsulation in amorphous boron nitride could suppress grain-boundary hydrogen loss [9].

Confidence: HIGH. The physics (large R_τ surplus) and the technique (pressure quench) are both demonstrated; the remaining challenge is engineering.

Prediction 2 (2030–2035): Room temperature at ambient pressure

For true RTSC ($T_c > 293$ K at 1 atm), the R_τ framework identifies three material families that could achieve $R_\tau(300\text{ K}) \geq 1$ through multi-channel pairing [9]:

(a) *Clathrate hydrides* (e.g., LaBeH₈-type, CaH₆ variants).—Computationally predicted $T_c > 300$ K under pressure. Materials with Debye temperature $\Theta_D > 1200$ K and electron-phonon coupling $\lambda > 2.5$ achieve $R_\tau > 1$ at 300 K via a single phonon channel. The challenge: retaining these parameters at ambient pressure.

(b) *Nickelate hydrides* (e.g., La₃Ni₂O₇+H).—Adding hydrogen to the bilayer nickelate introduces a fourth pairing channel. With three coherent channels each contributing $R_\tau^{(i)} \sim 0.4$, the multi-channel sum $R_\tau^{\text{multi}} = \sum_i R_\tau^{(i)} \sim 1.2$ exceeds unity [9].

(c) *Diamond-scaffold hydrides* (H:B:C:N).—Hydrogen and nitrogen co-doped boron-carbon diamond combines the highest known Θ_D (> 2000 K) with multiple dopant-induced pairing channels.

Confidence: MEDIUM. Requires discovery of new ambient-pressure materials or successful metastable stabilization of high-pressure phases.

Prediction 3 (2035–2045): Commercial RTSC products

Historical precedent sets the timescale from discovery to commercialization. YBa₂Cu₃O_{7- δ} (YBCO) was discovered in 1987 [8]; the first commercial YBCO wire entered the market in 1996—a 9-year gap [13]. MgB₂ ($T_c = 39$ K, discovered 2001) reached commercial wire by 2007 [14]. Both required solving engineering challenges: wire drawing, contact resistance, critical current density ($J_c > 10^4$ A/cm² for practical use), and upper critical field ($H_{c2} > 10$ T for magnets).

For RTSC, the absence of cryogenics removes the most expensive and fragile subsystem, but the material-specific challenges (grain-boundary weak links, anisotropy, brittleness) will still require ~ 10 years of engineering development. First products: RTSC power cables for short-distance urban grids and compact MRI magnets.

Confidence: MEDIUM. Depends on Predictions 1–2.

Prediction 4 (2040–2060): RTSC-enabled transformation

Systemic transformation—global lossless power grids, mass-produced personal maglev, portable MRI in every ambulance—requires not just an RTSC material but mature manufacturing at scale: kilometer-length wires, reliable joints, and $J_c > 10^5$ A/cm² in bulk form. This parallels the semiconductor industry’s trajectory from the

TABLE I. R_τ -based RTSC milestone timeline. R_τ values and material data from Ref. [9].

Milestone	Period	Key R_τ	Confidence
0 °C, ambient P	2027–2030	2.4–3.5	High
20 °C, ambient P	2030–2035	≥ 1.0	Medium
Commercial products	2035–2045	—	Medium
Systemic transformation	2040–2060	—	Low

transistor (1947) to the integrated circuit (1958) to the microprocessor (1971)—a 24-year arc.

Confidence: LOW. Long-range extrapolation with many intervening factors.

QUANTITATIVE IMPACT ANALYSIS

For each RTSC-enabled technology, we provide physics-grounded estimates. All cost figures are in 2025 US dollars.

Lossless power transmission

Global electricity transmission and distribution losses total ~ 2000 TWh/year, approximately 8% of total generation [15]. At a global average electricity price of $\sim \$0.10/\text{kWh}$, this represents $\sim \$200$ B/year in wasted energy. The carbon footprint of these losses, assuming the current global generation mix (60% fossil), corresponds to ~ 1.5 Gt CO₂/year—comparable to the entire aviation industry.

RTSC cables would eliminate resistive losses entirely. Residual losses in an AC system arise from dielectric hysteresis in cable insulation and eddy currents at joints, estimated at $< 0.1\%$ [16]. The R_τ requirement for year-round outdoor operation is stringent:

$$T_c > T_{\text{max}}^{\text{outdoor}} + \Delta T_{\text{margin}} = 50^\circ\text{C} + 10^\circ\text{C} = 333\text{ K}, \quad (3)$$

which requires $R_\tau(333\text{ K}) > 1$. For indoor cables (data centers, hospitals), $T_c > 293$ K suffices. The quenched-LaH₁₀ scenario ($T_c \sim 250$ K) would be suitable for underground cables (soil temperature ~ 10 – 15°C year-round) or refrigerated enclosures cooled to $< -20^\circ\text{C}$ —far cheaper than liquid nitrogen cooling of existing HTS cables.

Near-term opportunity: Urban power grids with underground cable runs at constant soil temperature ($\sim 15^\circ\text{C} = 288$ K) would benefit from any material with $T_c > 288$ K.

Compact MRI

Current clinical MRI scanners generate 1.5–3 T fields using NbTi wire cooled to 4.2 K in a liquid-helium bath. The cryogenic subsystem (cryostat, cold head, helium management) accounts for $\sim 40\%$ of system cost and $> 60\%$ of maintenance expenses [17]. A typical 1.5 T clinical MRI costs \$1.5–3 M, with annual operating costs of \$100–200 K (dominated by helium and electricity for the cold head).

An RTSC MRI would eliminate the entire cryogenic subsystem:

- Magnet cost: $\sim \$50$ K (wire + winding, no cryostat)
- Electronics: $\sim \$50$ – 100 K (gradient coils, RF, computing)
- Total system: $\sim \$100$ – 200 K
- Annual operating cost: $\sim \$10$ K (electricity only)

The R_T requirements are more demanding than for cables: $T_c > 300$ K, upper critical field $H_{c2} > 10$ T at operating temperature, and $J_c > 10^4$ A/cm². These are achievable for materials with $R_T \gg 1$, since large R_T correlates with large gap Δ and hence large $H_{c2} \propto \Delta^2/E_F$.

At \$150 K per unit, MRI becomes feasible for rural clinics, ambulances, and developing countries. The addressable market expands from $\sim 5,000$ units/year (current) to $\sim 50,000$ units/year, creating a $\sim \$7.5$ B/year new market. The public health impact—early detection of stroke, cancer, and traumatic brain injury in settings where CT is the only current option—is difficult to overstate.

Nuclear fusion

The magnetic confinement approach to fusion requires ~ 10 – 20 T fields over ~ 10 m³ volumes. ITER, the international fusion experiment, uses Nb₃Sn superconducting magnets cooled to 4.5 K at a project cost of $\sim \$25$ B [18]. The cryogenic plant alone costs $\sim \$1$ B and consumes ~ 30 MW—a significant parasitic load for a reactor targeting 500 MW fusion power.

The MIT-led SPARC/ARC design [19] already exploits high-temperature superconductors (HTS, ReBCO, $T_c \sim 90$ K) to achieve > 20 T fields in compact magnets, reducing reactor size by a factor of ~ 4 compared to ITER. RTSC magnets would provide two further advantages:

1. *No cryogenics*: eliminating the \$1 B cryogenic plant and its 30 MW parasitic load.

2. *Higher fields*: RTSC materials with large R_T are expected to have large gaps Δ and hence large H_{c2} , potentially enabling > 30 T fields that further shrink the plasma volume.

The net effect: fusion power plants $\sim 10\times$ smaller and $\sim 10\times$ cheaper than ITER-class devices. If RTSC materials are available by 2035, the timeline for commercial fusion power could accelerate from $\sim 2060+$ to ~ 2045 —a 15-year advance with enormous implications for decarbonization.

Superconducting magnetic energy storage

A persistent current in a superconducting loop stores energy indefinitely with zero resistive loss. Current superconducting magnetic energy storage (SMES) systems operate at 4 K and are limited to small scales (~ 1 – 10 MJ) by cryogenic cost [20].

RTSC SMES would eliminate cooling costs, enabling grid-scale deployment. A 1 MWh ($= 3.6$ GJ) RTSC solenoid with $B = 5$ T requires a coil volume

$$V = \frac{2\mu_0 E}{B^2} = \frac{2 \times 4\pi \times 10^{-7} \times 3.6 \times 10^9}{25} \approx 360 \text{ m}^3, \quad (4)$$

roughly the size of a standard shipping container (~ 33 – 76 m³ for 20–40 ft) when wound as a tightly packed toroid at 10 T (reducing volume by $4\times$). At 10 T and compact geometry, $V \sim 90$ m³—two standard containers.

Unlike lithium-ion batteries, SMES has no cycle degradation, no fire risk, and $> 95\%$ round-trip efficiency (limited only by the power electronics interface). RTSC SMES could replace lithium-ion for grid-scale storage applications where cycle life and safety are paramount.

Transportation: frictionless maglev

The Shanghai Transrapid maglev, operating since 2004, uses conventional superconducting magnets (NbTi at 4.2 K) and cost $\sim \$1.3$ B for 30 km of track [21]—\$43 M/km, compared to $\sim \$5$ – 15 M/km for conventional high-speed rail. The dominant cost driver is the cryogenic infrastructure embedded in every vehicle.

RTSC maglev eliminates vehicle-side cryogenics entirely. The track consists of passive RTSC rails carrying persistent currents; the vehicle carries permanent magnets (or vice versa). Estimated infrastructure cost reduction: 60–80%, to $\sim \$10$ – 15 M/km—competitive with conventional rail.

The physics is compelling: in the zero-drag limit, energy consumption is determined solely by aerodynamic resistance. For a maglev train at 300 km/h with $C_{DA} \approx 8$ m² and air density $\rho_{\text{air}} = 1.2$ kg/m³:

$$P_{\text{aero}} = \frac{1}{2} \rho_{\text{air}} C_{DA} v^3 \approx 3.3 \text{ MW}, \quad (5)$$

versus $\sim 8\text{--}12\text{ MW}$ for wheel-on-rail trains at the same speed (including rolling friction). Energy savings: $\sim 60\text{--}70\%$ per passenger-km.

Computing: superconducting interconnects

In modern data centers, $\sim 30\text{--}50\%$ of power consumption is attributable to resistive losses in wiring, power distribution, and cooling of the associated heat [22]. Global data center electricity consumption is $\sim 400\text{--}500\text{ TWh/year}$ (2024), implying $\sim 150\text{--}250\text{ TWh/year}$ dissipated as wiring heat.

RTSC interconnects would eliminate resistive wiring losses, reducing data center power consumption by $\sim 30\text{--}50\%$. At $\$0.08/\text{kWh}$ (industrial rate), this represents $\$12\text{--}20\text{ B/year}$ in savings.

Caveat: Superconductivity improves *wiring*, not *switching*. CMOS transistor operation does not require superconductivity; the logic gates themselves are non-superconducting. Josephson junction logic [23] could in principle replace CMOS, but its clock speeds ($\sim 100\text{ GHz}$, already demonstrated at 4 K) come with extremely low logic density. RTSC would make Josephson logic more accessible by removing cryogenics, but it would not solve the density problem.

WHAT RTSC CANNOT DO

Balanced assessment requires identifying technologies that will *not* benefit from RTSC:

(i) *Quantum computers.*—Superconducting qubits (transmons) operate at $\sim 15\text{ mK}$, not because they require superconducting wiring, but because qubit coherence demands thermal excitation energy $k_{\text{B}}T \ll \hbar\omega_q \sim 25\text{ }\mu\text{eV}$ ($\omega_q/2\pi \sim 5\text{ GHz}$). At room temperature ($k_{\text{B}}T \approx 25\text{ meV}$), thermal occupation $\bar{n} \sim 10^3$ destroys coherence regardless of whether the circuit is superconducting. RTSC would reduce cryogenic costs for *classical control electronics* but not eliminate the millikelvin stage.

(ii) *Battery charging speed.*—Charging rate is limited by electrochemical kinetics (lithium-ion diffusion in the cathode, SEI layer impedance), not wiring resistance. RTSC cables to the charger would reduce I^2R heating in the cable but not change the fundamental charge rate of the battery itself.

(iii) *Optical communication.*—Photons have zero electrical resistance by definition. Fiber-optic transmission losses ($\sim 0.2\text{ dB/km}$ at 1550 nm) arise from Rayleigh scattering and absorption, neither of which involves electrical resistance. RTSC is irrelevant to optical networks.

(iv) *Semiconductor logic speed.*—The clock speed of modern processors is limited by RC delay in interconnects and power dissipation in transistors, not by wire resistance alone. While RTSC eliminates the R in RC,

the capacitance C remains, and the switching energy per transistor (set by $\sim CV^2$) is unchanged.

THE R_{τ} ROADMAP

The path from the current state of the art to RTSC-enabled technology can be organized into three phases, each gated by the R_{τ} criterion:

Phase 1 (now–2030): Pressure-quench existing materials.— LaH_{10} ($R_{\tau} = 3.47$) and H_3S ($R_{\tau} = 1.89$) are the best candidates for kinetic trapping at ambient pressure. The key engineering challenge is rapid, controlled decompression at cryogenic temperatures. Success criterion: $T_{\text{c}} > 273\text{ K}$ at $P = 1\text{ atm}$.

Phase 2 (2030–2040): New ambient-pressure materials.—The three material families identified in Ref. [9]—clathrate hydrides, nickelate hydrides, and diamond-scaffold hydrides—are priority targets for first-principles screening and experimental synthesis. The open-source R_{τ} screener [26] enables community-driven search across the $> 150,000$ entries in the Materials Project database [12]. Success criterion: ambient-pressure material with $R_{\tau}(300\text{ K}) > 1$.

Phase 3 (2040+): Engineering at scale.—Wire drawing, joining, and critical current optimization for the winning material. Historical precedent (YBCO, MgB_2) suggests ~ 10 years from material discovery to first commercial products. Success criterion: $J_{\text{c}} > 10^5\text{ A/cm}^2$ in kilometer-length wires.

The role of community search

The R_{τ} criterion reduces the RTSC search to a tractable combinatorial problem: for each candidate material, compute Θ_{D} , λ , and the number of pairing channels from first-principles or database values, evaluate $R_{\tau}(300\text{ K})$, and rank. The R_{τ} screener code [26] implements this pipeline and is freely available. We encourage the materials science community—particularly groups with access to high-throughput DFT workflows [25]—to extend the screening beyond the 897 hydrogen-metallic compounds examined in Ref. [9] to the full inorganic materials space.

CONCLUSION

The R_{τ} framework, derived from quantum information theory, provides the first quantitative criterion for screening RTSC candidates and enables physics-grounded timeline predictions. The technological implications of RTSC are vast— $\$200\text{ B/year}$ in energy savings, universal MRI access, accelerated fusion, and frictionless transport—but they require materials with $T_{\text{c}} > 300\text{ K}$ at ambient

pressure, a goal that the R_τ analysis identifies as challenging but physically achievable.

The next experimental milestone is clear: pressure-quench LaH_{10} to ambient conditions and measure T_c . If the quenched material retains $T_c > 200$ K, the second electrical revolution will be underway.

The author thanks QuTech and the Quantum Inspire team for cloud access to the Tuna-9 processor, and the Materials Project team for the open database. This work was supported in part by the $\tau = 1 - F$ research program.

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- [1] H. K. Onnes, The resistance of pure mercury at helium temperatures, *Commun. Phys. Lab. Univ. Leiden* **12**, 120 (1911).
- [2] J. Bardeen, L. N. Cooper, and J. R. Schrieffer, Theory of superconductivity, *Phys. Rev.* **108**, 1175 (1957).
- [3] J. G. Bednorz and K. A. Müller, Possible high T_c superconductivity in the Ba-La-Cu-O system, *Z. Phys. B* **64**, 189 (1986).
- [4] M. Somayazulu *et al.*, Evidence for superconductivity above 260 K in lanthanum superhydride at megabar pressures, *Phys. Rev. Lett.* **122**, 027001 (2019).
- [5] A. P. Drozdov *et al.*, Superconductivity at 250 K in lanthanum hydride under high pressures, *Nature* **569**, 528 (2019).
- [6] A. P. Drozdov, M. I. Erements, I. A. Troyan, V. Ksenofontov, and S. I. Shylin, Conventional superconductivity at 203 kelvin at high pressures in the sulfur hydride system, *Nature* **525**, 73 (2015).
- [7] D. Deng and C. W. Chu, Pressure-quenched HgBa-CaCuO with $T_c = 151$ K at ambient pressure, Preprint (2026).
- [8] M. K. Wu *et al.*, Superconductivity at 93 K in a new mixed-phase Y-Ba-Cu-O compound system at ambient pressure, *Phys. Rev. Lett.* **58**, 908 (1987).
- [9] S.-K. Huang, Room-temperature superconductivity as a quantum information problem: The R_τ screening criterion and material predictions, Preprint (2026).
- [10] S.-K. Huang, Temporal asymmetry as Petz recovery failure, Zenodo DOI: 10.5281/zenodo.18897853 (2026); see also <https://github.com/akaiHuang/petz-recovery-unification>.
- [11] D. Petz, Sufficiency of channels over von Neumann algebras, *Q. J. Math.* **39**, 97 (1988).
- [12] A. Jain *et al.*, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, *APL Mater.* **1**, 011002 (2013).
- [13] D. Larbalestier, A. Gurevich, D. M. Feldmann, and A. Polyanskii, High- T_c superconducting materials for electric power applications, *Nature* **414**, 368 (2001).
- [14] M. Tomsic *et al.*, Overview of MgB_2 superconductor applications, *Int. J. Appl. Ceram. Technol.* **4**, 250 (2007).
- [15] International Energy Agency, *World Energy Outlook 2024* (IEA, Paris, 2024).
- [16] A. P. Malozemoff *et al.*, Progress in high temperature superconductor coated conductors and their applications, *Supercond. Sci. Technol.* **25**, 123001 (2012).
- [17] P. A. Rinck, *Magnetic Resonance in Medicine*, 12th ed. (ABW Wissenschaftsverlag, Berlin, 2018).
- [18] ITER Organization, *ITER Research Plan within the Staged Approach* (ITER Organization, St. Paul-lez-Durance, 2018).
- [19] A. J. Creely *et al.*, Overview of the SPARC tokamak, *J. Plasma Phys.* **86**, 865860502 (2020).
- [20] M. H. Ali, B. Wu, and R. A. Dougal, An overview of SMES applications in power and energy systems, *IEEE Trans. Sustain. Energy* **1**, 38 (2010).
- [21] R. Thornton, T. Clark, and E. Perreault, Linear synchronous motor propulsion of urban transit vehicles, in *Proceedings of the Joint Rail Conference* (ASME, 2006).
- [22] A. Shehabi *et al.*, *United States Data Center Energy Usage Report* (Lawrence Berkeley National Laboratory, 2016).
- [23] K. K. Likharev and V. K. Semenov, RSFQ logic/memory family: a new Josephson-junction technology for sub-terahertz-clock-frequency digital systems, *IEEE Trans. Appl. Supercond.* **1**, 3 (1991).
- [24] M. Gao *et al.*, Upper bound on the critical temperature of superconductors from first-principles phonon calculations, *Phys. Rev. B* **111**, 024506 (2025).
- [25] S. Curtarolo *et al.*, AFLOW: An automatic framework for high-throughput materials discovery, *Comput. Mater. Sci.* **58**, 218 (2012).
- [26] S.-K. Huang, R_τ screener: open-source code for superconductor screening, <https://github.com/akaiHuang/petz-recovery-unification> (2026).
- [27] S.-K. Huang, Measuring temporal asymmetry on a superconducting quantum processor: From quantum erasure to the birth of time's arrow, Preprint (2026).
- [28] A. Schilling, M. Cantoni, J. D. Guo, and H. R. Ott, Superconductivity above 130 K in the Hg-Ba-Ca-Cu-O system, *Nature* **363**, 56 (1993).